

Siavash Shami

M.Sc. in Geodesy | InSAR & Remote Sensing Researcher

Faculty of Geodesy and Geomatics Engineering

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About

Siavash Shami is a geodesist and InSAR researcher specializing in surface deformation monitoring, geohazards, and Earth Sciences. He received his M.Sc. in Geodesy from *K. N. Toosi University of Technology* in 2018, where his thesis focused on InSAR applications in open-pit mines. His research integrates InSAR, GNSS, and satellite remote sensing with geological and hydrological data to investigate natural and anthropogenic deformation, including earthquakes, landslides, salt-dome displacement, and infrastructure monitoring. He also applies machine learning and deep learning to deformation modeling and environmental remote sensing. A parallel focus of his research is on the technical and methodological aspects of InSAR, including error characterization, phase bias, atmospheric corrections, and interferogram network optimization, as well as the application of remote sensing data to study a wide range of Earth Sciences phenomena, from groundwater dynamics to land cover change and soil erosion. He has authored 7 peer-reviewed international journal articles published in journals indexed in Web of Science (WoS) across five major publishers: Elsevier, Springer Nature, IEEE, Taylor & Francis, and MDPI. He has also served as a peer reviewer for multiple WoS-indexed journals and has authored practical handbooks on InSAR processing.

Research Interests

- Radar, especially InSAR, imaging
- Geodesy and earth crustal deformation
- Satellite remote sensing
- AI for earth sciences applications
- Infrastructure monitoring, geohazard, & Climate-Driven land deformation

Educational Background

- M.Sc. in Geodesy

K. N. Toosi University of Technology, Tehran, Iran | 2018

Thesis: *A Feasibility Study on the Application of InSAR for Surveying in Open Pit Mines (In Persian)*
- B.Sc. in Surveying Engineering

Eyn Ol Ghozat Higher Education Institute, Mianeh, Iran | 2016

Experiences

Scientific Reviewer for publications of	Elsevier: Gondwana Research (1), Advances in Space Research (1), Remote Sensing Applications: Society and Environment (3) IEEE: IEEE Access (1) Springer Nature: Earth Science Informatics (1) Tech Science Press: Revue Internationale de Géomatique (1) Others: Geography, environment, sustainability (1)
Awards and Honors	Ranked 1st among B.Sc. students of Surveying Engineering (GPA: 3.93/4.00) 2016
Research & Professional Experiences	Research Collaboration - InSAR Applications 2023–2024 Collaborated with Dr. Meisam Amani on InSAR-based investigations of landslides in New Zealand, dam deformation and shoreline changes in Canada, & open-pit mine deformation in South Africa. Methods & Tools: Sentinel-1, PS/SBAS, GMTSAR, StaMPS, ISCE, MintPy, QGIS InSAR & GIS Consultant/Research Collaborator - Bonyan Zamin Co. 2022 Prepared annual InSAR-based surface displacement maps of Isfahan, Iran, for Isfahan Municipality, using Sentinel-1 data, PS-InSAR, GMTSAR, StaMPS, and QGIS

Compulsory Military Service | 2019-2021

Served as a full-time conscript; during this period, he continued independent research and technical writing and authored a practical handbook on InSAR processing

Skills

InSAR	Category I: Technical InSAR SAR Data: Sentinel-1, ENVISAT, ALOS-1/2, ERS-1/2 Interferometric Processing Methods: PS-InSAR, SBAS, NSBAS, DInSAR Interferometric Processing & Phase Unwrapping: Multilooking & noise reduction, Interferogram Network Design & Optimization Error Sources & Corrections: Atmospheric correction (ERA5, GACOS, spatial-temporal filtering), Phase-bias identification & correction InSAR Software & Processing Platforms: StaMPS/MTI, GMTSAR, MintPy, ISCE, LiCSBAS, SNAP, DORIS, ROI_PAC Category II: Applications & Physical Interpretation Natural Hazards & Geological Deformation: Landslide detection, earthquakes, Fault & Salt-dome deformation Hydrology & Environmental Deformation: Groundwater-related deformation & land subsidence, Geological–hydrological interpretation of surface deformation, Coastal deformation & coastline displacement Engineering & Infrastructure Monitoring: Railways subsidence, Dams deformation monitoring, Urban areas displacement & land subsidence, Open-pit mines displacement & deformation monitoring
Geodesy & Physical Surveying	Fields: Satellite Geodesy, Physical Geodesy, Levelling, Topographic Surveying Surveying Instrumentation: Total Station, Nivo, Dual-Frequency GNSS receivers Software: Civil 3D, Global Mapper, Google Earth, RTKLIB
Remote Sensing & Earth Observation	Hydrological & Water Resources: Soil moisture (Landsat, Sentinel-1), Groundwater storage (GRACE, piezometric well data), Precipitation (CHIRPS, TRMM) Land Surface & Environmental Monitoring: Land-cover/land-use change (Sentinel-1/2, MODIS, Landsat) Air Quality & Climate change: Air pollution (Sentinel-5P/TROPOMI, MODIS AOD), Climate classification & trend analysis (Köppen–Geiger, long-term climate data) Soil & Geomorphology: Soil erosion modeling (RUSLE) Platform & Tools: Google Earth Engine
Computational & Analytical Skills	Artificial Intelligence & Data Science Machine Learning: Random Forest, XGBoost, SVM, Linear/Lasso Regression Deep Learning: RNN/LSTM for time-series forecasting Programming & Scripting Python: NumPy, Pandas, Matplotlib Linux: Ubuntu GIS & Spatial Analysis QGIS: Spatial analysis, raster/vector processing, cartography
Languages	Azerbaijani: Native Persian: Native English: Professional working proficiency

Publications

17. Karami, E., **Shami, S.**, Maghsoudi, Y., Ranjgar, B., & Azadnejad, S. (2025). Investigating the InSAR Phase Bias in the SBAS Algorithm and Its effect on different Landcovers. *IEEE Access* (peer-reviewed: Research Article) doi.org/10.1109/ACCESS.2025.3568144

16. Naboureh, A., Li, A., Bian, J., Lei, G., Nan, X., Zhang, Z., **Shami, S.** & Lin, X. (2024). Green space coverage versus air pollution: a cloud-based remote sensing data analysis in Sichuan, Western China. *International Journal of Digital Earth* (peer-reviewed: Research Article) doi.org/10.1080/17538947.2024.2383454

15. **Shami, S.**, Shahriari, M. A., Nilfouroushan, F., Forghani, N., Salimi, M., & Reshadi, M. A. M. (2024). Surface displacement measurement and modeling of the Shah-Gheyb salt dome in southern Iran using InSAR and machine learning techniques. *International Journal of Applied Earth Observation and Geoinformation* (peer-reviewed: Research Article)
doi.org/10.1016/j.jag.2024.104016
14. Abdalla, A., **Shami, S.**, Shahriari, M. A., & Azar, M. K. (2024). Estimation of land displacement in East Baton Rouge Parish, Louisiana, using InSAR: Comparisons with GNSS and machine learning models. *The Egyptian Journal of Remote Sensing and Space Sciences* (peer-reviewed: Research Article)
doi.org/10.1016/j.ejrs.2024.02.008
Co-first author (equal contribution)
13. Khoshlahjeh Azar, M., **Shami, S.**, Nilfouroushan, F., Salimi, M., Ghayoor Bolorfroshan, M., & Reshadi, M. A. M. (2022). Integrated analysis of Hashtgerd plain deformation, using Sentinel-1 SAR, geological and hydrological data. *Scientific Reports* (peer-reviewed: Research Article)
doi.org/10.1038/s41598-022-25659-4
Co-first author (equal contribution)
12. **Shami, S.**, Azar, M. K., Nilfouroushan, F., Salimi, M., & Reshadi, M. A. M. (2022). Assessments of ground subsidence along the railway in the Kashan plain, Iran, using Sentinel-1 data and NSBAS algorithm. *International Journal of Applied Earth Observation and Geoinformation* (peer-reviewed: Research Article)
doi.org/10.1016/j.jag.2022.102898
11. **Shami, S.**, Ranjgar, B., Bian, J., Khoshlahjeh Azar, M., Moghimi, A., Amani, M., & Naboureh, A. (2022). Trends of CO and NO2 Pollutants in Iran during COVID-19 pandemic using Timeseries Sentinel-5 images in Google Earth Engine. *Pollutants* (peer-reviewed: Research Article)
doi.org/10.3390/pollutants2020012
10. **Shami, S.** (2021). GMTSAR installation and processing Guide - Practical InSAR handbook series. *ebook* (Book: Handbook)
doi.org/10.5281/zenodo.22609191
9. **Shami, S.**, Khoshlahjeh, M., Ghorbani, Z., Moghimi, A., Mohammadzadeh, A., & Sabet Ghadam, S. S. (2021). Evaluation of air pollution contributes for the COVID-19 pandemic in Iran using Sentinel 5 Satellite Data. *Journal of Geomatics Science and Technology* (In Persian) (peer-reviewed: Research Article)
8. Hossainali, MM., & **Shami, S.** (2020). Processing Radar Images using StaMPS. *K.N. TOOSI UNIVERSITY Press* (In Persian, ISBN: 978-622-6655-46-0) (Book: Technical Book)
Co-first author (equal contribution)
7. **Shami, S.**, & Ghorbani, Z. (2019). Processing Radar Images using GMTSAR & SNAP. *Arshadan Press* (In Persian, ISBN: 978-622-251-021-3) (Book: Technical Book)
6. **Shami, S.**, & Ghorbani, Z. (2019). Investigating water storage changes in Iran using grace and chirps data in the google earth engine system. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* (Conference Paper: Poster Presentation)
doi.org/10.5194/isprs-archives-XLII-4-W18-981-2019
5. **Shami, S.**, & Ghorbani, Z. (2019). Influence of wavelength radar images in estimation of open-pit mine displacements using radar interferometry technology (case study: Gol Gohar Sirjan mine). *3rd TRIGGER International Conference* (Conference Paper: Poster Presentation)
4. **Shami, S.**, Hossainali, MM., & Babaee, S. (2019). Analysis of Large-scale Displacement Using Radar Interferometry Technology in Open-pit Mines (Case Study: Gol Gohar Sirjan Mine). *Geospatial Engineering Journal* (In Persian) (peer-reviewed: Review Article)
3. **Shami, S.**, Ghorbani, Z., & Abedi, L. (2018). Calculation of the extracted mines volume using DEM obtained from the Envisat radar satellite images (Case study: Gol Gohar Sirjan Mine). *The 4th National Geology and Mining Explorations Symposium* (In Persian) (Conference Paper: Oral Presentation)
2. **Shami, S.**, Ghorbani, Z., & Abedi, L. (2018). Persistent Scatterers Time series analysis of radar interferometry to determine the rate subsidence of areas around the open pit mines (Case study: Gol Gohar Sirjan Mine). *The 4th National Geology and Mining Explorations Symposium* (In Persian)

(Conference Paper: Poster Presentation)

1. **Shami, S.**, & Hossainali, MM. (2018). Analysis of the effect of temporal and spatial baseline of radar images on the amount of coherence for detecting large-scale displacement gradients using interferometric synthetic aperture radar (Case study: Gol Gohar Sirjan Mine). Geomatics and GIT 97 (In Persian) (Conference Paper: Poster Presentation)